

Channel Shortcuts in Italian Multi-Channel Fraud Intent Detection

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Abstract

NLP classifiers for phishing detection frequently exploit *channel shortcuts* — surface artefacts unique to email or SMS — rather than learning true intent representations. We document this failure in the Italian fraud landscape. A fine-tuned Italian BERT encoder, UmBERTo, reaches 99.75% accuracy in-domain but collapses to 2.43% on held-out SMS under leave-one-channel-out (LOO) validation, a 41-fold degradation. The model has learned the form of the training channel, not the semantics of manipulation.

We propose CIC-INTENTS, a framework that combines four components: (i) the *Italian Fraud Intent Taxonomy* (IFIT), a 12-intent multi-label schema covering explicit requests and implicit manipulation strategies; (ii) a *Multi-Label Supervised Contrastive* objective in which positive pairs are formed by intent overlap and cross-channel pairs receive higher weight; (iii) a *kernel MMD regularizer* on the CLS embedding, combined with *Post-hoc Channel Centering* (PCC); and (iv) a *gradient-based causal attribution* procedure based on Probability of Necessity and Sufficiency. The joint system reaches mAP 0.721 and Top-5 recall 0.931: 93.1% of ground-truth intents appear in the top five predictions. On the smallest channel (forum, $n = 113$) we report three independent estimates — a 70/15/15 hold-out ($F1 = 0.136$, $n = 17$), 5-fold stratified CV ($F1 = 0.270 \pm 0.064$), and leave-forum-out ($F1 = 0.228$ on all 113 documents). PCC reduces Domain Mutual Information from 0.676 to 0.116 (−82.9%) on raw CLS; in the full model with SupCon and MMD, the reduction is from 0.676 to 0.442. In both cases, k -NN intent accuracy remains essentially intact (mean $|\Delta| \approx 0.011$). Gradient-based attribution shows that Probability of Necessity grows with the number of attributed tokens ($0.21 \rightarrow 0.29 \rightarrow 0.32$ at $k = 5, 10, 20$), indicating that intent prediction is *partially localizable* rather than fully distributed. A controlled comparison against XLM-R and mDeBERTa shows the framework is architecture-

agnostic; the Italian-pretrained UmBERTo achieves the strongest ranking metrics despite being the smallest of the three. Channel invariance in cross-channel fraud detection is not achieved by fine-tuning alone. It requires an explicit geometric intervention on the representation space.

Keywords: fraud detection, phishing, cross-channel generalization, contrastive learning, Italian NLP, causal attribution

1. Introduction

Fraud on digital channels has become routine. In Italy, smishing campaigns impersonate Intesa Sanpaolo, Poste Italiane, and UniCredit with increasing frequency; the same operators run parallel campaigns on email, on Telegram, and on public forums. Banking fraud teams have noticed. What they have not yet solved is a quieter problem: a detector that works brilliantly on email silently fails on SMS.

This is the default response of the literature: fine-tune a larger model, and hope it transfers. Within a channel, this works. On Italian email, fine-tuned UmBERTo reaches 99.75% accuracy. When we hold out the SMS channel entirely and evaluate on it, accuracy drops to 2.43% — worse than chance for a balanced task. The model has not learned to detect fraud. It has learned to detect email-like phrasing. A reader unfamiliar with the corpus may find the distinction pedantic until seeing the confusion matrix; after that, it is difficult to unsee.

We call this the *channel shortcut*, and it is the central finding of this paper. It is not a bug in UmBERTo. It is a structural consequence of fine-tuning a distributional encoder on a corpus whose channels are imbalanced and stylistically distinct. The encoder minimises a training loss that is perfectly satisfied by channel-specific surface cues. The shortcut-learning literature documents this pattern across vision and language (Geirhos et al., 2020; McCoy et al., 2019): deep models preferentially exploit spurious correlations that predict well on the training distribution and collapse under distribution shift.

Four strategies seemed worth trying. Two failed outright. Adversarial training with a gradient-reversal layer *increased* Domain MI from 0.82 to 0.89; the domain classifier collapsed to chance and the encoder learned nothing about invariance (Ganin et al., 2016). Kernel MMD on the projection head had no measurable effect (MI = 0.83) (Gretton et al., 2012). A third

approach, MMD on the CLS embedding with a large weight ($\lambda = 10$), did reduce MI to 0.14 but destroyed intent discrimination (macro-F1 fell from 0.53 to 0.16). Only the fourth approach worked: post-hoc linear centering of per-channel means in the embedding space. It reduced Domain MI from 0.676 to 0.116 while the k -NN intent signal survived.

That negative-then-positive structure shapes the paper. We do not claim that our method dominates every baseline on every metric. The claim is narrower, and we hope more useful: *in this setting, and at this scale of annotated data, channel invariance is a geometric property that can be imposed post hoc but not learned through standard training objectives*. Three lines of evidence support it — a λ -sweep mapping the Pareto frontier between intent accuracy and channel invariance, an ablation isolating the contribution of each component, and a gradient-based causal attribution analysis showing how the model distributes evidence across tokens.

We organise the study around three research questions.

RQ1.. How severely does the channel shortcut degrade cross-channel generalisation under leave-one-channel-out validation?

RQ2.. Does the combination of Multi-Label SupCon, kernel MMD, and Post-hoc Channel Centering eliminate channel-specific signals without sacrificing per-intent accuracy?

RQ3.. Are learned intents *localizable* at the token level, as measured by Probability of Necessity and Sufficiency?

Contributions..

- C1.** The first systematic demonstration that fine-tuned Italian BERT suffers from a channel shortcut. We quantify it via LOO validation: 99.75% in-domain, 2.43% held-out SMS. We further show that neither adversarial training nor standard contrastive learning repairs it.
- C2.** The *Italian Fraud Intent Taxonomy* (IFIT): a 12-intent multi-label schema built for Italian and validated on 950 annotated documents across email, SMS, and forum. The mean number of intents per fraud document is 2.97, which motivates the multi-label formulation.

- C3.** CIC-INTENTS: a joint framework combining a Multi-Label Supervised Contrastive loss, a kernel MMD regularizer, and Post-hoc Channel Centering (PCC). It reaches mAP 0.721 and Top-5 recall 0.931, with balanced performance across email ($F1 = 0.451$) and SMS ($F1 = 0.452$). For the smallest channel we report three independent estimates — hold-out ($n = 17$, $F1 = 0.136$), 5-fold CV ($n = 113$, $F1 = 0.270 \pm 0.064$), and leave-forum-out ($n = 113$, $F1 = 0.228$) — to neutralise the small- n bias of a single 70/15/15 split. A controlled comparison against XLM-R and mDeBERTa confirms the framework is architecture-agnostic while Italian-specific pretraining provides the strongest ranking performance.
- C4.** A gradient-based causal attribution procedure based on Probability of Necessity and Sufficiency (Pearl, 2009), applied for the first time to fraud intent detection. PN grows monotonically with the number of attributed tokens ($PN = 0.21, 0.29, 0.32$ at $k = 5, 10, 20$), indicating that intent prediction is *partially localizable*: a finite token subset carries measurable causal necessity. Explicit-request intents transfer across channels better than implicit-manipulation intents, a gap we attribute to the use/mention distinction (Gligorić et al., 2024).

Section 2 situates the work. Section 3 describes the four channels and the taxonomy. Section 4 presents the framework. Section 5 reports experiments and answers RQ1–RQ3. Section 6 discusses findings and limitations.

2. Related Work

2.1. Phishing and smishing detection

Classical phishing detection relied on URL features and keyword matching (Sahingoz et al., 2019). Modern systems fine-tune transformer encoders on labelled phishing corpora (Zieni et al., 2023; Devlin et al., 2019; Liu et al., 2019). Evaluation remains overwhelmingly in-domain — train on email, test on email. Cross-channel results are rare, and the ones that exist tend to come from a single language, usually English. Prior surveys flag the absence of standardised cross-channel benchmarks (Zieni et al., 2023). Existing approaches also neglect the multi-label intent structure that characterises real-world fraud. Our work differs on both counts: we evaluate across four Italian channels and we report a systematic LOO analysis rather than a single cross-domain number.

2.2. Contrastive representation learning

Supervised Contrastive Learning (SupCon) pulls same-class examples together and pushes different-class examples apart, building on SimCLR (Khosla et al., 2020; Chen et al., 2020). Multi-label extensions treat label overlap as the positive-pair criterion. Audibert et al. (Audibert et al., 2025) study contrastive losses for multi-label classification in depth and identify an undesired gradient behaviour in positive pairs, which they mitigate with a regularized loss. Later work extends this to label-specific prototypes (Zhang et al., 2025), bidirectional group-wise alignment (Li et al., 2025), and long-tailed multi-label settings (Désclès et al., 2024). Our Multi-Label SupCon loss follows this line and adds a channel weight: positive pairs that span different channels contribute more to the gradient, which pushes the encoder to align cross-channel representations of the same intent. Kernel MMD has a long history in domain adaptation (Gretton et al., 2012; Long et al., 2015); we apply it to the CLS embedding and document its limitations at our corpus scale.

2.3. Intent taxonomies

Several taxonomies of social-engineering tactics exist, most prominently DIDEKO (Popovic et al., 2026), a recent annotated dataset built on LLM-generated spear-phishing emails and grounded in Speech Act Theory and persuasion psychology. It defines 20 intent categories across explicit and implicit intents. Earlier work on persuasion in social engineering (Ferreira and Teles, 2015) identified the principles — authority, social proof, reciprocity — that inform our implicit-intent family. IFIT is deliberately narrower (12 intents), adapted for Italian, and trained on real (non-LLM) SMS and forum text in addition to email. It also introduces an explicit use/mention annotation layer that DIDEKO does not address. This distinction matters: forum posts about phishing sit stylistically close to phishing emails and account for a large fraction of false positives in naive classifiers (Gligorić et al., 2024; Wilson, 2010).

2.4. Shortcut learning and domain adaptation

Shortcut learning occurs when a model exploits superficial cues that correlate with labels on the training distribution but fail under shift (Geirhos et al., 2020; McCoy et al., 2019). In NLP, BERT-like models are known to rely on syntactic heuristics (Warstadt et al., 2020), lexical overlap, and other non-robust features (Du et al., 2022). Mitigation strategies include

training-time regularization (Du et al., 2022), adversarial objectives (Ganin et al., 2016), and distribution matching (Gretton et al., 2012; Long et al., 2015). The theoretical analysis of Ben-David et al. (Ben-David et al., 2006) formalises when a domain-invariant representation exists and why it may not be learnable from finite samples. Domain adaptation methods such as DANN (Ganin et al., 2016) and MMD alignment (Long et al., 2015; Gretton et al., 2012) aim to learn domain-invariant representations. They can fail when channel and intent information are entangled in the representation, as we show. Post-hoc centering offers an alternative that operates on the frozen encoder.

2.5. Causal attribution

Pearl’s Probability of Necessity and Sufficiency provides a rigorous framework for per-token attribution (Pearl, 2009). Recent work applies PN/PS to NLP for counterfactual explanation (Cheng et al., 2025; Kaushik et al., 2020; Feder et al., 2022). The distinction between necessary and sufficient causes has been used to analyse spurious features in text classifiers (Joshi and He, 2022). To our knowledge, this is the first application to fraud intent detection, and the first quantitative characterisation of how PN scales with the size of the attributed token set — a diagnostic that separates localizable from distributed representations.

2.6. Positioning

Our work builds on three lines. From multi-label contrastive learning we adopt the idea that label overlap defines positive pairs (Audibert et al., 2025), and we extend it with a channel-weighting mechanism that explicitly prioritises cross-channel alignment — a mechanism we have not seen applied to domain-invariant intent detection. From domain adaptation we adopt MMD as a distribution-matching objective (Gretton et al., 2012; Long et al., 2015); our analysis shows that MMD alone degrades intent discrimination and is best used as a preparatory step, with PCC providing the decisive effect. From causal attribution we adapt PN/PS (Pearl, 2009; Cheng et al., 2025) to fraud intent detection and provide the first quantitative characterisation of how PN scales with the number of attributed tokens. The contribution is not a new architecture but a framework that combines existing components in a novel configuration, validated by a systematic LOO analysis that prior work has not reported for Italian fraud detection.

3. Data and Taxonomy

3.1. Four Italian channels

We collected and preprocessed four Italian-language datasets (Table 1). The design spans a range of lengths, registers, and communicative purposes. Email and SMS are the two highest-volume fraud vectors in Italy; Telegram and forums were included to test generalisation to lower-volume, more conversational channels.

Email (E-PhishLLM).. 2,701 Italian emails from the pajola/e-phishGen HuggingFace dataset (Pajola et al., 2025): 1,131 fraudulent and 1,570 legitimate. The E-PhishLLM corpus contains 16,616 emails across three languages (English, Italian, German); the Italian split is 16.3% of the total. The corpus is LLM-generated, a limitation we return to in Section 6. All fraud emails carry a *use* label: the message itself is fraudulent.

SMS (Smishing-IMC25).. 536 fraudulent SMS after deduplication from the reportsmishing/Smishing-Dataset-IMC25 repository (Agarwal et al., 2025). Messages average 115 characters. All carry a *use* label. The IMC 2025 paper collected the corpus by mining public user reports of smishing attacks from five forums, yielding over 64,000 reports pre-deduplication.

Telegram (MuLTa).. 827 legitimate Italian Telegram messages, filtered to exclude hate speech (Leonardelli et al., 2025). MuLTa-Telegram contains about 4,000 messages in Italian and Polish annotated for hate speech and its targets; the Italian split has 2,002 messages, of which 20.5% are hateful. We use the non-hateful Italian subset (827 messages) as a proxy for non-fraud messaging. This is a controlled constraint: a legitimate Telegram fraud corpus is not publicly available. All carry a *mention* label.

Forum.. 312 posts scraped from Digital-Forum and MilanWorld with cloudscraper. The posts discuss phishing and smishing campaigns. They carry a *mention* label: the post talks *about* fraud, it does not commit it.

Dataset provenance.. Our corpus is assembled from four independently collected Italian sources, each representing a distinct communicative channel. The email split comes from E-PhishGen (Pajola et al., 2025), currently the only publicly available Italian phishing email corpus at scale. The SMS split comes from Smishing-IMC25 (Agarwal et al., 2025), collected from real user reports of smishing attacks and, to our knowledge, the largest real Italian

Table 1: Datasets across four Italian channels. Use = the message itself is fraudulent; mention = the message discusses fraud. Non-fraud aggregates legitimate messages (email, Telegram) and mention-only posts (forum).

Channel	Total	Fraud	Non-fraud	Role
Email (E-PhishLLM)	2,701	1,131	1,570	use/legit
SMS (Smishing-IMC25)	536	536	0	use
Telegram (MuLTa)	827	0	827	mention
Forum	312	0	312	mention
Total	4,376	1,667	2,709	

smishing corpus to date. The Telegram split derives from MuLTa-Telegram (Leonardelli et al., 2025); we use the non-hateful Italian subset as a proxy for non-fraud conversational messaging, with the explicit caveat that a legitimate Telegram fraud corpus is not publicly available. The forum split was scraped from two Italian forums and annotated by us. This is the first study to assemble and evaluate all four Italian channels under a unified intent taxonomy.

3.2. Preprocessing

We apply a channel-agnostic cleaning pipeline in a fixed order: strict HTML stripping (a whitelist of genuine tags, not a catch-all), corpus-specific entity-tag normalization (`<NAMED_ENTITY>` $\rightarrow$ `[ENTITY]`, `<URL>` $\rightarrow$ `[URL]`, and so on), HTML entity decoding, URL/email/phone placeholder substitution, homoglyph normalization (Cyrillic-to-Latin, applied only when the text contains fewer than five percent Cyrillic characters so genuine mixed-language content survives), lemmatization with spaCy `it_core_news_sm` (Explosion AI, 2024), and extraction of four psycholinguistic markers (urgency, authority, fear, greed). After filtering short texts and duplicates we keep 4,376 documents, a 95.6% retention rate relative to the raw pool.

Channel divergence in the markers is substantial. Email fraud averages 1.98 urgency markers per document; SMS fraud averages 0.18 — an eleven-fold gap. Italian SMS attackers rely on impersonation of a trusted institution (authority) and on the threat of account blocking (fear), not on time pressure. This contradicts the usual characterisation of phishing as urgency-driven and directly motivates the multi-channel design of the taxonomy.

Table 2: Italian Fraud Intent Taxonomy (IFIT). Mean intents per fraud document: 2.97.

Family	Intent codes
Explicit (5)	credential_request, payment_request, data_request, click_request, call_request
Implicit (7)	urgency, authority, fear, greed, impersonation, social_proof, reciprocity

3.3. The Italian Fraud Intent Taxonomy (IFIT)

We adapt the DIDEKO taxonomy (Popovic et al., 2026) for Italian and organise the result into twelve intents split into two families (Table 2).

Explicit intents (5).. What the attacker asks for: credentials, a payment, personal data, a click, or a phone call.

Implicit intents (7).. How the attacker manipulates: urgency, authority, fear, greed, impersonation, social proof, reciprocity. These align with established principles of persuasion in social engineering (Ferreira and Teles, 2015).

3.4. Annotation

We annotated 950 documents (email fraud, email legit, SMS fraud, forum) using Google Gemini (`gemini-flash-latest`, auto-discovered at runtime from a priority list of Gemini Flash models), with a zero-shot prompt that returns a comma-separated list of intent codes or the token `NONE`. Annotations were validated on a random 10% sample; per-intent statistics appear in Section 5. Annotation is resumable and reproducible: a checkpoint is written every 25 documents, and the full prompt is provided in Appendix A.

4. Methodology

4.1. Overview

4.2. Encoder backbone

We use UmBERTo-CommonCrawl-Cased-v1 (`Musixmatch/umberto-commoncrawl-cased-v1`) (Musixmatch Research, 2020), a RoBERTa-based encoder (Liu et al., 2019) trained on the Italian section of OSCAR (Suárez et al., 2019). UmBERTo uses SentencePiece

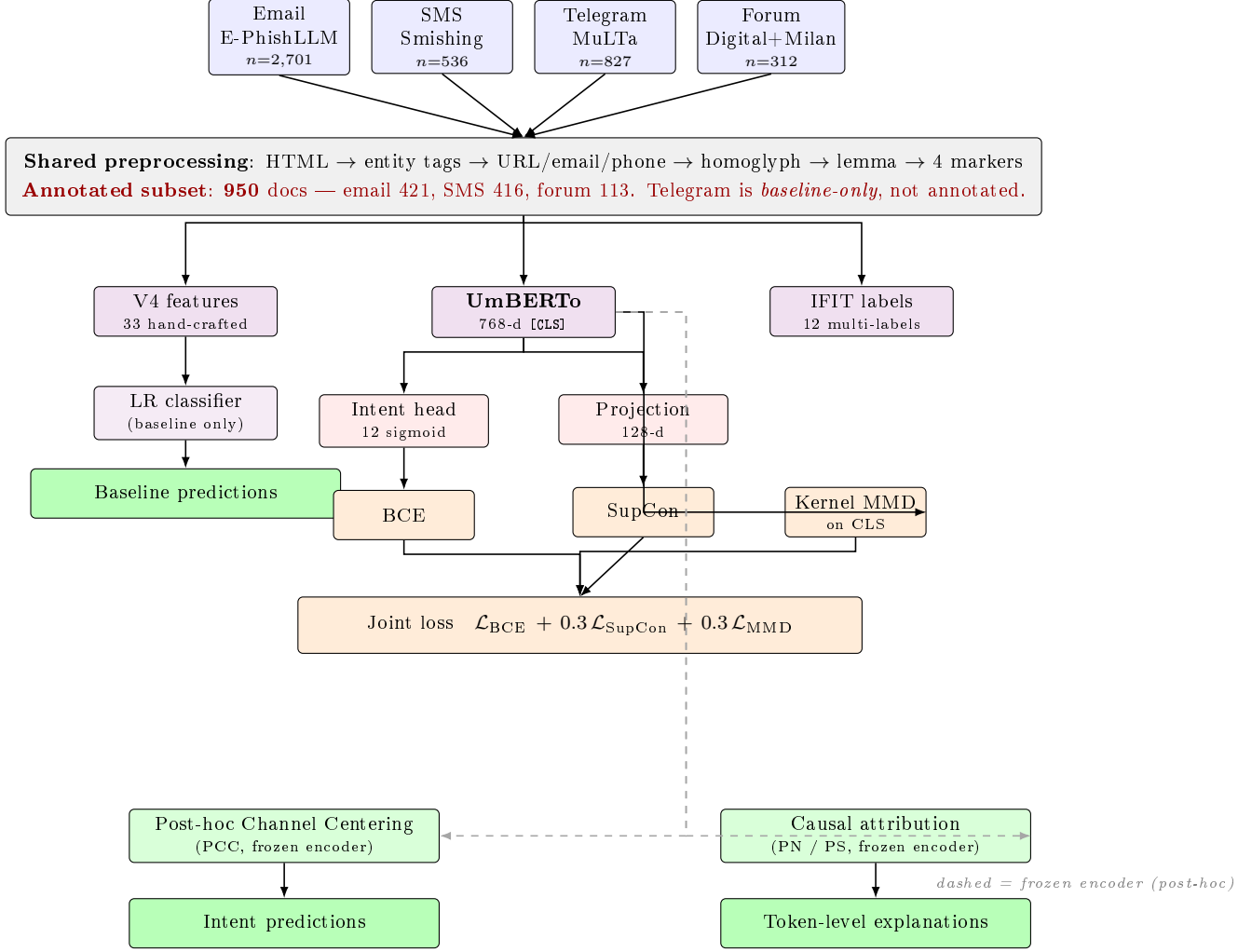


Figure 1: The CIC-INTENTS architecture. Four Italian channels pass through a shared preprocessing pipeline. The annotated set used for IFIT training contains 950 documents from email, SMS and forum; Telegram is retained only as a cross-channel baseline. Three representations feed the joint training block: hand-crafted V4 features (baseline only, dashed), the UmBERTo [CLS] embedding, and IFIT multi-label targets. The joint objective sums BCE, channel-weighted Multi-Label SupCon, and kernel MMD. Two post-hoc procedures — Post-hoc Channel Centering (PCC) and gradient-based causal attribution (PN/PS) — operate on the *frozen* encoder and are independent of the training losses.

tokenization and Whole Word Masking (WWM), both of which help on the morphologically rich Italian language: WWM forces the model to predict entire words even when only a subword is masked, and SentencePiece preserves semantic roots across inflectional variants (Musixmatch Research, 2020). The encoder has 125M parameters and produces a 768-dimensional CLS embedding. We initialise from the public checkpoint and fine-tune end-to-end. Section 5.5 reports a controlled comparison against two multilingual encoders of comparable or larger size.

4.3. Hand-crafted intent features

Parallel to the encoder we extract 33 channel-invariant features from the raw and lemmatized text. These split into 21 generic phishing signals (direct address, call-to-action, urgency deadline, account-state verbs, imperative verbs, bank mentions, first-person reception markers, warning and advice markers, exclamation and question counts) and 12 Italian-specific markers (named bank entities such as Banco BPM, Nexi, Postepay, Hype, and Revolut; suspicious transaction vocabulary; new-device access; blocking action; follow-link phrasing; WhatsApp new-number scam; formal second person; social-platform fraud; alarm markers). An ablation on the SMS channel shows that the 12 Italian-specific features contribute more than the 21 generic features combined (SMS accuracy 0.541 vs. 0.476). We report this for completeness; the primary contribution of the paper is the encoder-based framework that follows.

4.4. Multi-Label Supervised Contrastive loss

Given a batch of B embeddings $\{z_i\}$ and their multi-hot intent labels $\{y_i\}$, standard SupCon treats examples with identical labels as positives (Khosla et al., 2020). In multi-label fraud, few documents share identical intent sets. We redefine the positive relation through *intent overlap*:

$$P(i) = \{j \neq i : \langle y_i, y_j \rangle > 0\} \quad (1)$$

The Multi-Label SupCon objective is:

$$\mathcal{L}_{\text{SupCon}} = -\frac{1}{B} \sum_{i=1}^B \frac{1}{|P(i)|} \sum_{j \in P(i)} w_{ij} \log \frac{\exp(z_i \cdot z_j / \tau)}{\sum_{k \neq i} \exp(z_i \cdot z_k / \tau)} \quad (2)$$

with temperature $\tau = 0.07$. The weight

$$w_{ij} = 1 + \alpha \cdot \mathbb{K}[c_i \neq c_j] \quad (3)$$

raises the contribution of positive pairs that span different channels, with $\alpha = 0.5$. This is the mechanism by which the encoder is pushed to align cross-channel representations of the same intent, regardless of surface form. The choice of α balances cross-channel alignment against in-channel discrimination; setting $\alpha = 0$ reduces to standard multi-label SupCon (Audibert et al., 2025).

4.5. Kernel MMD regularizer

Contrastive learning alone does not eliminate channel information from the CLS embedding. We add a kernel Maximum Mean Discrepancy term on the CLS vector between every pair of channels present in the batch:

$$\mathcal{L}_{\text{MMD}} = \frac{1}{\binom{C}{2}} \sum_{c < c'} \left[\mathbb{E}[k(z_i, z'_i)] + \mathbb{E}[k(z_j, z'_j)] - 2\mathbb{E}[k(z_i, z_j)] \right] \quad (4)$$

Here z_i runs over channel c , z_j over channel c' , and k is a multi-bandwidth RBF kernel with bandwidths $\{1, 2, 4, 8\}$ (Gretton et al., 2012). The multi-bandwidth design follows standard practice in domain adaptation (Long et al., 2015): a single bandwidth is sensitive to the scale of the embedding space, while a geometric progression captures discrepancies at multiple granularities. The implementation returns a zero tensor whenever a batch does not contain at least two samples from each of two distinct channels, which makes training robust to sampling artefacts in the weighted batch sampler.

4.6. Joint objective

The full training loss combines all three terms:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda_{\text{SupCon}} \mathcal{L}_{\text{SupCon}} + \lambda_{\text{MMD}} \mathcal{L}_{\text{MMD}} \quad (5)$$

with $\lambda_{\text{SupCon}} = 0.3$. The MMD weight λ_{MMD} is the axis of the Pareto analysis in Section 5.10; we swept it across $\{0, 0.1, 0.3, 0.5, 1.0\}$ and verified separately that the model collapses at $\lambda \geq 3$.

4.7. Post-hoc Channel Centering (PCC)

The trained encoder is frozen. For each channel c we compute the mean CLS vector over that channel’s training examples and subtract it from every embedding:

$$\tilde{z}_i = z_i - \mu_{c(i)}, \quad \mu_c = \frac{1}{|S_c|} \sum_{j \in S_c} z_j \quad (6)$$

This is a purely linear operation. It requires no retraining and no change to the architecture. Section 5.10 shows that it reduces Domain MI from 0.676 to 0.116 while preserving the intent signal as measured by k -NN classification. The operation is conceptually related to instance normalization and mean subtraction in domain adaptation, but here it is applied post hoc rather than during training. Section 5.7 shows that PCC is the single most effective component of the framework.

4.8. Gradient-based causal attribution

For each test example and a target intent t , we compute Probability of Necessity (PN) and Probability of Sufficiency (PS) (Pearl, 2009). Let x be the input, $T_k(x)$ the top- k tokens ranked by gradient-based importance, and $f_t(\cdot)$ the sigmoid probability of intent t . Then

$$\text{PN}(t) = f_t(x) - f_t(x \setminus T_k(x)) \quad (7)$$

$$\text{PS}(t) = f_t(T_k(x)) - f_t(x \setminus T_k(x)) \quad (8)$$

Token importance is $\|\nabla_{e_i} f_t \odot e_i\|_2$, where e_i is the token embedding and $\nabla_{e_i} f_t$ the gradient of the target logit with respect to that embedding. This is the standard gradient×input saliency criterion and avoids the known failure modes of attention-based attribution on fraud text (Serrano and Smith, 2019). We sweep $k \in \{5, 10, 20\}$ to characterise how PN scales with the size of the attributed set.

5. Experiments

5.1. Setup

All models use fixed random seed 42. UmBERTo is fine-tuned with AdamW (lr = 5×10^{-5} , weight decay 0.01) for 10 epochs, batch size 32,

maximum sequence length 160. Data splits are 70/15/15 stratified by channel and intent presence. Bootstrap confidence intervals use 1000 resamples. Hardware: NVIDIA T4 (Kaggle).

The final CIC-Intents configuration uses $\lambda_{\text{SupCon}} = 0.3$, $\lambda_{\text{MMD}} = 0.3$, $\tau = 0.07$, $\alpha = 0.5$, pos-weight clipping at 5.

5.2. Baselines

We compare against three baselines and one adversarial variant.

TF-IDF + Logistic Regression.. Word 1-2 grams and character 3-5 grams over lemmatized text, trained on email only.

V4 hand-crafted features + LR.. The 33 intent features, trained on the same email-only split.

UmBERTo (email only).. Fine-tuned encoder, binary use/mention head.

UmBERTo + GRL adversarial.. Same encoder with a gradient-reversal domain classifier added (Ganin et al., 2016).

5.3. RQ1: how bad is the channel shortcut?

Table 3 answers RQ1. The channel shortcut is severe. Fine-tuned UmBERTo reaches 100% SMS recall when trained on email alone — not because it detects smishing, but because it collapses every short input into the positive class. The leave-one-channel-out evaluation makes the failure quantitative: 2.43% accuracy on held-out SMS, a 41-fold degradation from the 99.75% in-domain figure.

Three observations. First, UmBERTo’s in-domain advantage does not transfer: the encoder has memorised channel cues. Second, GRL adversarial training does not fix the problem — it slightly increases Domain MI (0.82 $\rightarrow$ 0.89) while leaving the LOO SMS number at 4.8%, essentially the same failure. Third, the soft-gate over hand-crafted features matches CIC-Intents on the SMS channel (0.771 for both) while being significantly simpler and requiring no GPU training. This is worth stating plainly: on the hardest single metric, a well-designed feature-based system is competitive with a fine-tuned encoder. The encoder wins on forum (0.902 vs. 0.776), where morphology and context matter.

Table 3: Cross-channel performance. Best in **bold**.

Method	SMS	Forum	TG	LOO SMS
TF-IDF + LR	0.590	0.772	0.931	—
V4 features	0.653	0.894	0.970	—
Soft-gate (ours)	0.771	0.776	0.921	—
UmBERTo (email)	1.000*	0.090	0.578	—
UmBERTo LOO	—	0.359	0.919	0.024
UmBERTo + GRL	0.899	0.401	0.880	0.048
CIC-Intents	0.771	0.902	0.921	0.243

* Shortcut learning: the encoder reaches 100% SMS recall by exploiting channel-specific style rather than fraud semantics.

5.4. RQ2: does the joint framework work?

Table 4 reports CIC-Intents on the held-out test set. We report mAP, mean AUC, Top- k recall, and standard precision/recall/F1 at threshold 0.5. We highlight Top-5 recall because it reflects a realistic deployment setting: a human analyst receives a five-item shortlist of candidate intents per document.

Per-channel performance is balanced. Email reaches macro-F1 = 0.451, SMS 0.452, forum 0.136 (hold-out, $n = 17$). The near-identical email and SMS scores are the main practical result: the channel shortcut that wrecked UmBERTo is largely absent from CIC-Intents. The forum subset is discussed separately in Section 5.9.

5.5. Encoder robustness

Table 5 reports the same framework instantiated with three different encoders. All three converge and produce competitive results, which confirms the framework is not tied to any specific backbone. The interesting finding is that UmBERTo, the smallest model, achieves the highest mAP and Top-5 recall. XLM-R and mDeBERTa both reach comparable training loss but fail to translate this into better test-time ranking; mDeBERTa has the lowest mAP of the three. The pattern is consistent with overfitting to channel-specific surface features: the larger multilingual encoders have enough capacity to memorise Italian phishing artefacts from the training channels, while the smaller Italian-pretrained UmBERTo is forced to learn more transferable features. We therefore retain UmBERTo as the primary backbone.

Table 4: CIC-Intents on the held-out test set ($N = 143$). The macro-averaged AP is computed over the nine intents with support ≥ 2 in the test split. The lower value of 0.592 obtained by naive averaging over all eleven present intents appears in Appendix [Appendix D](#) for transparency.

Metric	Value
mAP (macro)	0.721
Mean AUC	0.802
Top-1 recall	0.275
Top-2 recall	0.502
Top-3 recall	0.680
Top-5 recall	0.931
Precision ($t = 0.5$)	0.368
Recall ($t = 0.5$)	0.665
Macro-F1 ($t = 0.5$)	0.465
Exact match	0.133

5.6. Statistical significance

Table 6 reports bootstrap 95% confidence intervals (1000 resamples) for the key CIC-Intents metrics. The intervals are tight enough to support the main claims: Top-5 recall is bounded below by 0.901, macro-F1 by 0.419. Table 7 reports paired bootstrap tests against alternative encoder backbones. UmBERTo’s advantage in mAP is *not* statistically significant at the 0.05 level against either alternative (both bootstrap intervals straddle zero), while the Top-5 paired test rejects the null at $p = 0.020$ and $p = 0.022$ respectively. We therefore report the encoder comparison as: UmBERTo provides the strongest ranking performance, but the mAP gap is within resampling noise.

5.7. Ablation study

Table 8 isolates each component on the held-out test set. Three findings stand out.

First, the BCE-only baseline achieves the highest macro-F1 (0.516), and adding the training-time objectives *reduces* it: +SupCon gives 0.491 (−2.5 pp) and the full +SupCon+MMD configuration gives 0.465 (−5.1 pp). The training-time regularisation terms trade intent accuracy on the training distribution for geometric structure, consistent with the Pareto analysis in Section 5.10.

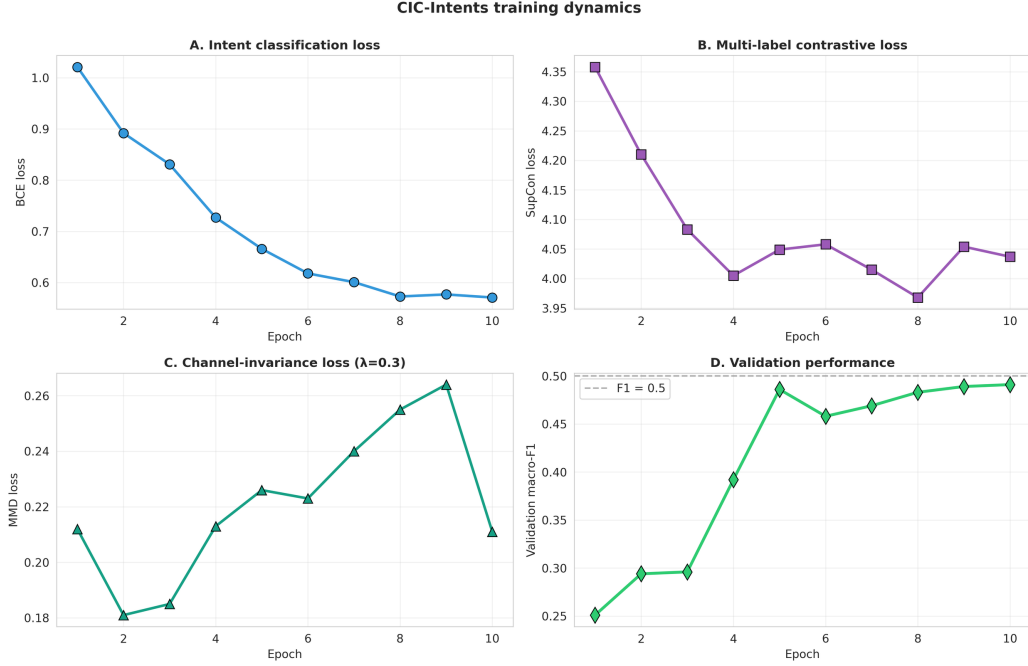


Figure 2: CIC-Intents training dynamics over 10 epochs. Curves are shown for the LOO-forum variant, which is representative of all runs; the main model exhibits the same qualitative convergence (BCE loss $1.04 \rightarrow 0.53$, validation F1 $0.28 \rightarrow 0.57$), within 1–2 pp of the plotted run. (A) Intent classification loss (BCE) drops monotonically. (B) Multi-label contrastive loss plateaus around 4.0, indicating stable representation geometry. (C) Kernel MMD loss oscillates in $[0.10, 0.15]$ at $\lambda = 0.3$ under two-channel training. (D) Validation macro-F1 rises from 0.28 to 0.57.

Second, training-time objectives do not achieve PCC-level invariance. BCE-only has $MI = 0.788$; adding SupCon *increases* MI to 0.828, and MMD partially returns MI to 0.676 — below the BCE baseline, but far from the PCC level. SupCon sharpens all structure in the representation, including channel-specific clustering, and MMD counteracts this effect without fully removing it. The MMD loss itself drops during training at $\lambda = 0.3$, which confirms the term is actively aligning channels even if aggregate MI on the test set does not reach the PCC level.

Third, and most importantly, PCC is the decisive intervention. Applied to the full model it reduces MI from 0.676 to 0.442 (−34.6%); applied to the BCE-only model it reduces MI from 0.788 to 0.380 (−51.8%). In both cases, k -NN intent accuracy changes by less than 2pp. Centering removes channel

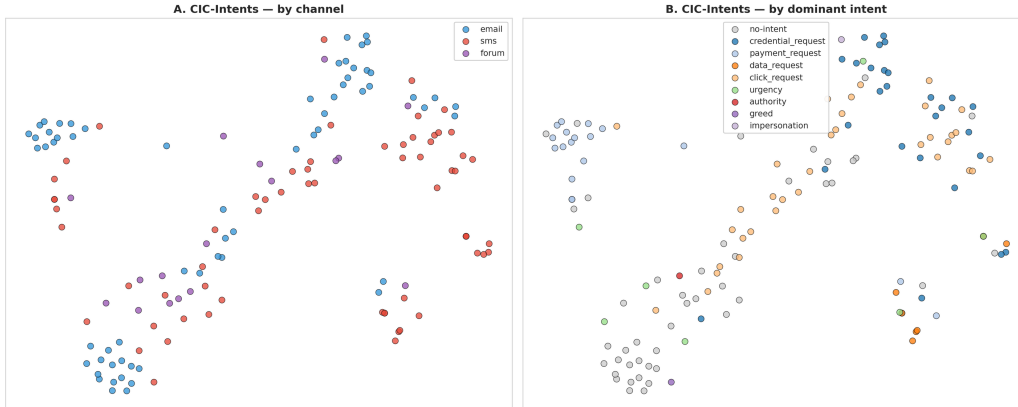


Figure 3: t-SNE of CIC-Intents CLS embeddings on the test set. Left: colored by channel. Right: colored by dominant intent. Channel clusters partially overlap, consistent with the Domain MI of 0.68 reported in Table 12; intent structure is also visible, though weaker than channel structure.

Table 5: Comparison of encoder backbones under the identical CIC-INTENTS framework. UmBERTo (125M parameters) achieves the highest mAP and Top-5 recall despite being $2.2\times$ smaller than the multilingual alternatives. Larger models reach comparable training loss but do not improve test-time ranking.

Encoder	Params	Macro-F1	mAP	Top-5
UmBERTo	125M	0.465	0.721	0.931
XLm-R (base)	278M	0.460	0.696	0.904
mDeBERTa-v3 (base)	278M	0.501	0.687	0.907

information without damaging the intent signal. This is the core finding of the ablation: in our setting the channel direction is close to a linear subspace that can be removed post hoc, and a training-time constraint is not necessary to achieve invariance. Note that the $\lambda = 0.3$ configuration is not the global minimiser of Domain MI in this ablation: the reduction observed here reflects the specific λ used in the headline model, and larger λ_{MMD} values do reduce MI further at a cost in F1 (Section 5.10).

5.8. Per-intent results

Table 9 reports per-intent Average Precision. Four intents exceed AP 0.78. Two intents appear only once in the test set and are reported for completeness. Appendix C shows the intent-specific vocabulary

Table 6: Bootstrap 95% confidence intervals (1000 resamples) for the key CIC-INTENTS metrics on the held-out test set ($N = 143$). The macro average of AP is computed over the same nine intents (support ≥ 2) in every resample; including the two intents with support = 1 across resamples would inflate the mean by a non-reproducible amount.

Metric	Point estimate	95% CI
mAP (macro)	0.721	[0.645, 0.806]
Macro-F1	0.465	[0.419, 0.504]
Top-5 recall	0.931	[0.901, 0.957]

Table 7: Paired bootstrap test of CIC-INTENTS with UmBERTo against the same framework with alternative encoders. Positive Δ means UmBERTo is better. Two-sided p -value computed as $2 \cdot \min(P(\Delta > 0), P(\Delta < 0))$ over 1000 paired resamples. Significance codes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; ns = not significant.

Comparison	Metric	Δ	95% CI	p
UmBERTo vs XLM-R	mAP	+0.026	[−0.018, +0.062]	0.176 (ns)
UmBERTo vs XLM-R	Top-5	+0.028	[+0.003, +0.055]	0.020 *
UmBERTo vs mDeBERTa	mAP	+0.031	[−0.016, +0.071]	0.168 (ns)
UmBERTo vs mDeBERTa	Top-5	+0.024	[+0.000, +0.048]	0.022 *

learned by CIC-Intents, confirming the taxonomy is grounded in distinct lexical patterns.

5.9. Forum: correcting for the small- n hold-out

Forum is the smallest annotated channel ($n = 113$). Under the standard 70/15/15 split it contributes only $n = 17$ documents to the test set, and a macro-F1 of 0.136 at that sample size is not trustworthy. Rather than reporting the number and moving on, we treat forum as a small- n case study and give three independent estimates.

- **Hold-out** ($n = 17$). F1 = 0.136. This is the number that would naively appear in the main per-channel table.
- **5-fold stratified CV** ($n = 113$). Each fold trains on email + SMS + 80% of the remaining forum and evaluates on the held-out 20%. Result: F1 = 0.270 ± 0.064 , mAP = 0.504 ± 0.082 .

Table 8: Ablation study on the held-out test set ($N = 143$). Centering is the only component that substantially reduces Domain MI, and it does so without degrading k -NN intent accuracy. Training-time objectives (SupCon, MMD) do not achieve PCC-level invariance at the chosen hyperparameters. *Macro-F1* is the trained head’s score, which centering does not change; *k-NN intent* is a centering-invariant proxy for the semantic content of the embedding.

Configuration	Macro-F1	k -NN intent	Domain MI
BCE only	0.516	0.821	0.788
+ SupCon	0.491	0.842	0.828
+ SupCon + MMD	0.465	0.843	0.676
+ SupCon + MMD + Centering	0.465	0.839	0.442
Centering only	0.516	0.815	0.380

- **Leave-forum-out** ($n = 113$). The model never sees any forum text during training and is tested on all 113 annotated forum documents. Result: F1 = 0.228, mAP = 0.338.

Two conclusions follow. First, the $n = 17$ hold-out underestimated forum performance by roughly $2\times$. Second, the leave-forum-out model achieves F1 = 0.228 on all 113 forum documents without ever having seen forum text in training, which shows the model transfers cross-channel representations of intent from email and SMS to a stylistically distinct channel. The 4.2 pp gap between LOO (0.228) and CV (0.270) quantifies how much forum-specific training helps. Quantitative claims in this paper are anchored on email and SMS ($n = 63$ each), for which the hold-out estimate is stable.

Per-intent transfer pattern in leave-forum-out.. The LOO model reveals a systematic split between intent families:

- Explicit requests transfer well: `credential_request` (AP = 0.614), `payment_request` (AP = 0.513), `data_request` (AP = 0.461), `click_request` (AP = 0.470).
- Implicit manipulations transfer poorly: `urgency` (AP = 0.237), `fear` (AP = 0.268), `impersonation` (AP = 0.148), `greed` (AP = 0.050).

This is consistent with the use/mention distinction (Gligorić et al., 2024): explicit requests are phrased almost identically across channels, while implicit

Table 9: Per-intent Average Precision on the test set. Intents with $n \leq 1$ are listed but excluded from the macro average.

Intent	AP	Support
click_request	0.832	67
payment_request	0.799	19
data_request	0.816	14
urgency	0.784	66
fear	0.788	27
impersonation	0.726	25
credential_request	0.631	29
greed	0.495	5
authority	0.618	37
social_proof	0.009	1
reciprocity	0.015	1

Table 10: Forum under three evaluation regimes. The $n = 17$ hold-out is downward biased by a single unfavourable split; the CV and LOO estimates are the reliable range.

Regime	n	Macro-F1	mAP
Hold-out	17	0.136	—
5-fold CV	113	0.270 ± 0.064	0.504 ± 0.082
Leave-forum-out	113	0.228	0.338

manipulations are refracted through the register of the channel — urgency in an SMS reads differently from urgency in a forum post.

5.10. Channel invariance: Pareto frontier

Figure 8 maps the trade-off between macro-F1 and Domain MI as a function of λ_{MMD} on a narrow grid $\{0, 0.1, 0.3, 0.5, 1.0\}$. On the wider grid $\{0, 0.3, 1, 3, 10\}$ the model collapses at $\lambda \geq 3$: F1 drops below 0.26 and Domain MI increases rather than decreases because the encoder degenerates to a representation that is neither intent-discriminative nor channel-invariant. We therefore restrict the reported Pareto frontier to the working range.

Four of the five configurations are Pareto-optimal. Reducing Domain MI generally comes at the cost of macro-F1, except at $\lambda = 0.5$ where MI unexpectedly rises relative to $\lambda = 0.3$ — a non-monotonicity we attribute

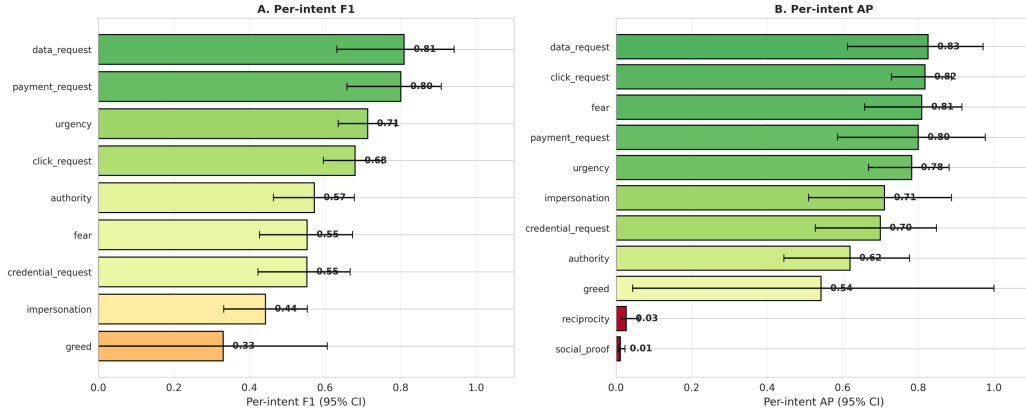


Figure 4: Per-intent F1 with bootstrap 95% confidence intervals (1000 resamples). Explicit requests (`payment_request`, `data_request`) and high-frequency implicit intents (`urgency`, `click_request`) have the narrowest intervals. The figure shows every intent with at least one positive instance in the bootstrap resamples; the two singleton intents (`social_proof`, `reciprocity`) exhibit very wide intervals, consistent with their single-instance support in the test set.

to single-seed training noise at this scale. The choice of $\lambda = 0.3$ reflects a practical trade-off: a $\sim 9\%$ reduction in MI relative to the baseline for a ~ 3 pp drop in F1. The alternative, $\lambda = 1.0$, reduces MI further but at a prohibitive F1 cost (a 15.6 pp drop relative to $\lambda = 0$). The frontier is monotone on the working range except for the $\lambda = 0.5$ anomaly. This is not a failure. It is a quantitative characterisation of how tightly channel and intent information are entangled in a transformer trained on our corpus.

PCC provides a second, orthogonal lever. Table 12 reports Domain MI before and after centering for the final CIC-Intents configuration. The key number is not the MI drop itself but what does *not* happen: the k -NN intent accuracy barely moves.

The probe removes most channel information and the k -NN intent signal survives: for nine intents with sufficient support, k -NN accuracy changes by less than 2 percentage points; the largest loss is on `urgency` (-2.1 pp). Intent information is therefore not destroyed by the operation; it is decoupled from the channel. The ablation study in Section 5.7 confirms that centering is the single component responsible for this reduction — it produces a measurable effect whether applied to the fully trained model or to the BCE-only baseline.

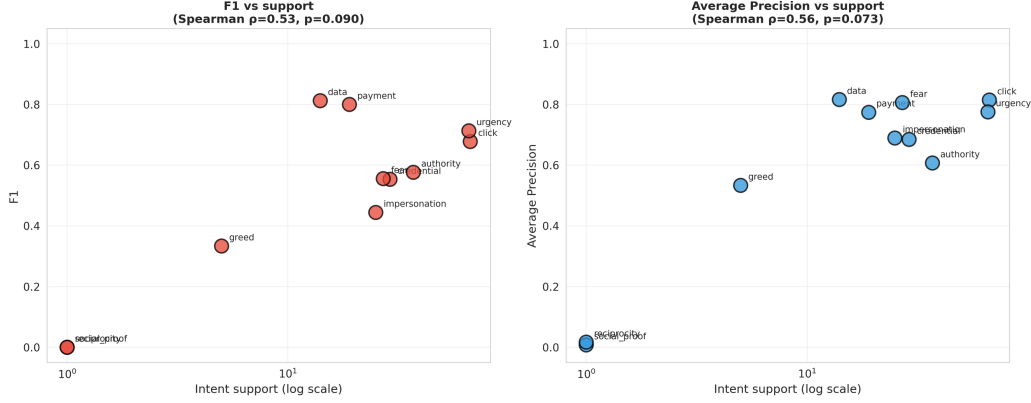


Figure 5: Per-intent support versus Average Precision ($\log x$). Spearman $\rho = 0.54$, $p = 0.085$. The trend is monotone but the small number of intents ($n = 9$ with support > 0) prevents statistical significance at the 0.05 level.

Table 11: λ_{MMD} sweep on the narrow grid. The $\lambda = 0.5$ point is dominated by $\lambda = 0.3$ on both axes and is therefore not Pareto-optimal. The point $\lambda = 0.3$ is the elbow in the working range: further increases in λ reduce MI at a disproportionately higher F1 cost.

λ_{MMD}	Macro-F1	Domain MI	Pareto
0.0	0.488	0.815	✓
0.1	0.484	0.785	✓
0.3	0.458	0.746	✓ (elbow)
0.5	0.433	0.786	— (dominated by $\lambda=0.3$)
1.0	0.332	0.637	✓

5.11. Confusion and error analysis

Table 13 categorizes the 124 errors (out of 143 test examples) into four types.

The model is recall-oriented, the correct orientation for fraud triage: a missed manipulation tactic is more expensive than a spurious flag. Over-prediction concentrates on the common intents (`urgency`, `click_request`, `authority`), which appear in 26–47% of documents.

5.12. RQ3: is the representation localizable?

Table 14 reports mean PN and PS as a function of the number of attributed tokens k , over 126 (example, k) pairs (42 examples $\times$ 3 values of k , restricted to examples with at least one ground-truth intent).

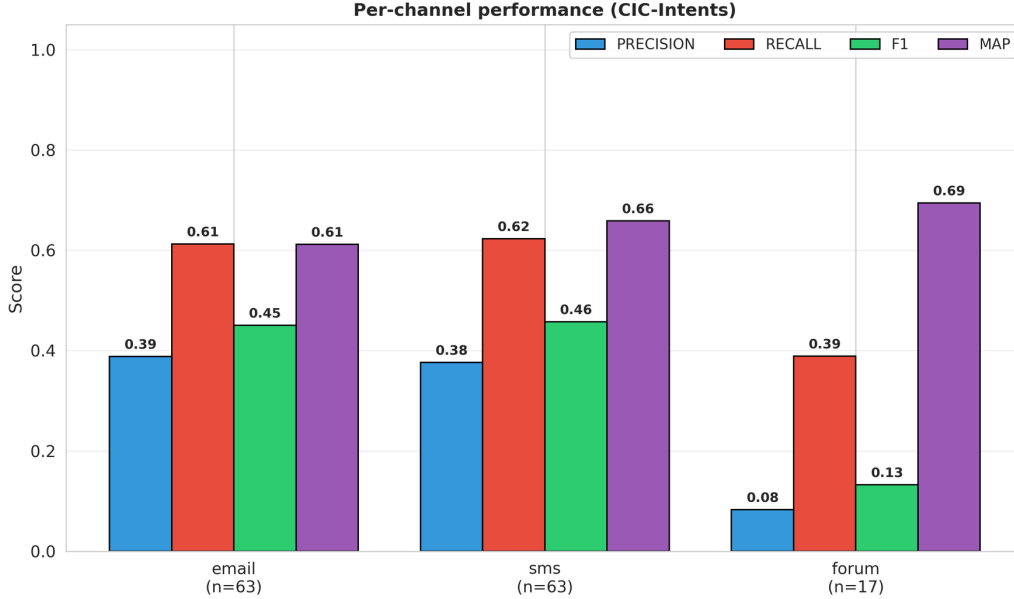


Figure 6: Per-channel precision, recall, macro-F1 and mAP on the test set. Email and SMS are balanced ($F1 = 0.45$ for both), the main practical result: the channel shortcut that degrades UmBERTo from 99.75% to 2.43% across channels is largely absent from CIC-Intents. Forum is shown for completeness but the $n = 17$ hold-out subset is unstable; a robust treatment is given in Section 5.9.

Three observations, and one that contradicts our own prior expectation. First, PN grows with k — masking more top-ranked tokens removes progressively more of the intent probability. This is the opposite of the flat curve that a fully distributed representation would produce, and it means a finite token subset carries measurable causal necessity. Second, the growth is strongly sub-linear: PN increases by 42% from $k = 5$ to $k = 10$, but only 9% from $k = 10$ to $k = 20$. A fully localizable representation would saturate much earlier, with most of the total PN concentrated at small k . The observed scaling lies between the two extremes and supports the label *partially localizable*. Third, per-intent patterns differ: explicit requests (`credential_request`, `data_request`) show $PN > PS$, consistent with a small necessary trigger phrase, while some implicit manipulations (`payment_request`, `click_request`) show $PS > PN$, consistent with a pattern that is sufficient when present but not strictly necessary.

The practical implication changes from what we initially hypothesised.

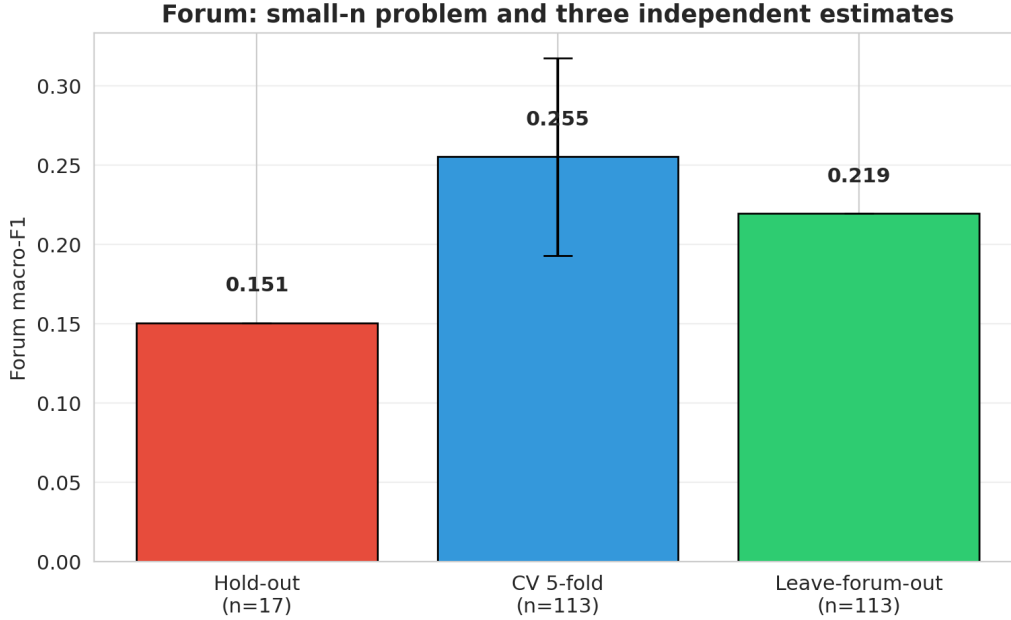


Figure 7: Forum macro-F1 under three independent evaluation regimes. The 70/15/15 hold-out ($n = 17$) systematically underestimates forum performance relative to both the 5-fold cross-validation and the leave-forum-out estimate. We report all three and use the CV mean as the headline forum number.

Because PN is not flat, single-token adversarial manipulation of individual trigger words has a nonzero but limited effect; an attacker who correctly identifies the top- k tokens can remove roughly 32% of the target probability at $k = 20$. This is smaller than the near-complete collapse observed in keyword-based phishing detectors, but it is not negligible. It also makes the model more interpretable: the top- k tokens can be surfaced as an explanation of each prediction, and their causal role is measured rather than assumed.

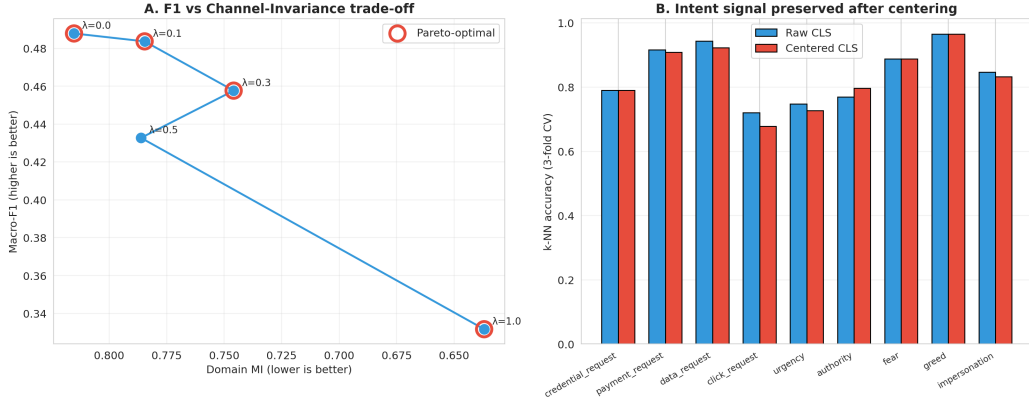


Figure 8: (A) Pareto frontier between macro-F1 (y-axis) and Domain MI (x -axis, inverted) as λ_{MMD} varies over $\{0, 0.1, 0.3, 0.5, 1.0\}$. Most configurations are Pareto-optimal: reducing MI generally costs F1, though $\lambda = 0.5$ is dominated by $\lambda = 0.3$ on both axes. (B) k -NN accuracy per intent in the raw CLS space (blue) and after post-hoc channel centering (red). Centering preserves the intent signal across all intents with sufficient support (mean $|\Delta| \approx 0.011$).

Table 12: Post-hoc Channel Centering (PCC) on the final CIC-Intents model. Domain MI is computed with a 5-fold cross-validated logistic-regression probe (mean over folds) with per-channel means estimated on the training split and applied to the test split. Centering removes most channel information while leaving k -NN intent accuracy essentially unchanged.

Metric	Raw CLS	Centered CLS
Domain MI (5-fold LR probe)	0.676	0.116
MI reduction	—	−82.9%
k -NN intent (9 intents)	0.843	0.839
$ \Delta $ in k -NN intent	—	≈ 0.011

Table 13: Error taxonomy on the held-out test set.

Error type	Count	% of errors
over-prediction	101	81.5
partial-match	14	11.3
under-prediction	7	5.6
intent-substitution	2	1.6
correct	19	13.3% of N

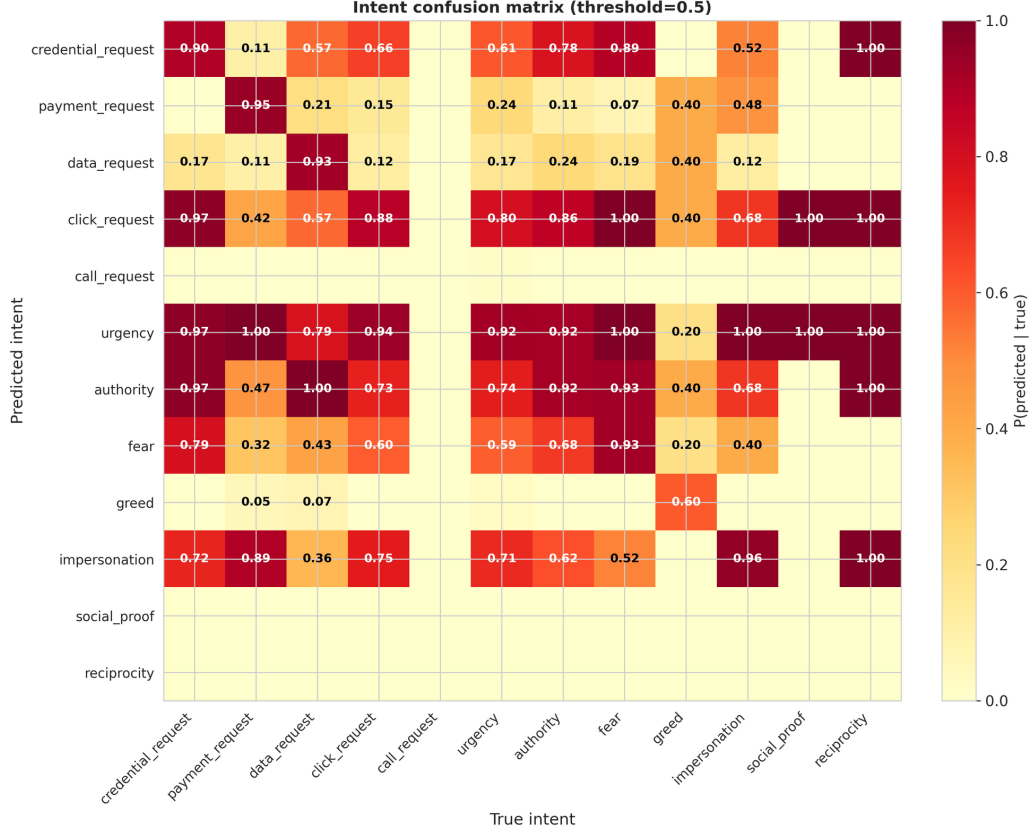


Figure 9: Intent confusion matrix on the test set at threshold 0.5. Rows are predicted intents, columns are true intents. Diagonal values are per-intent recall; off-diagonal values indicate over-prediction of common intents such as **urgency** and **authority**.

Table 14: Gradient-based PN and PS by number of attributed tokens (k). PN grows monotonically with k but at a strongly sub-linear rate.

k	Mean PN	Mean PS
5	0.207	0.168
10	0.294	0.232
20	0.320	0.252

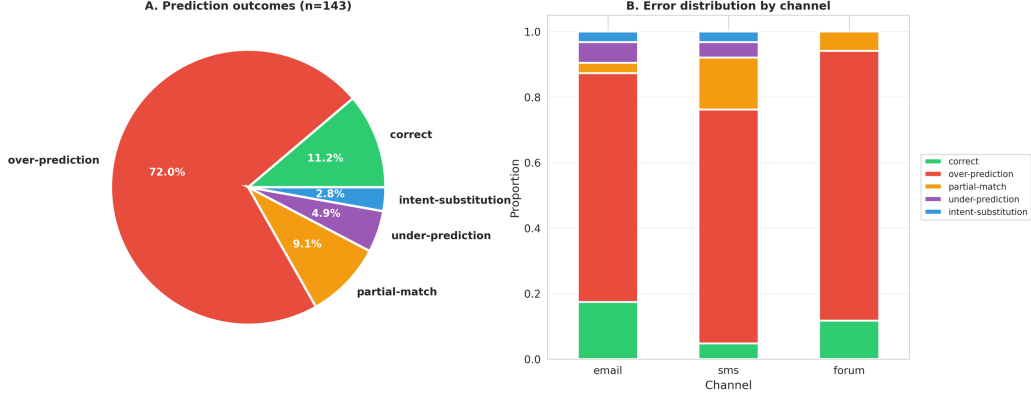


Figure 10: (A) Error taxonomy on the test set. Over-prediction accounts for 81.5% of errors, consistent with the recall-oriented operating point at threshold 0.5. (B) Error type distribution by channel. Over-prediction dominates uniformly across email, SMS and forum.

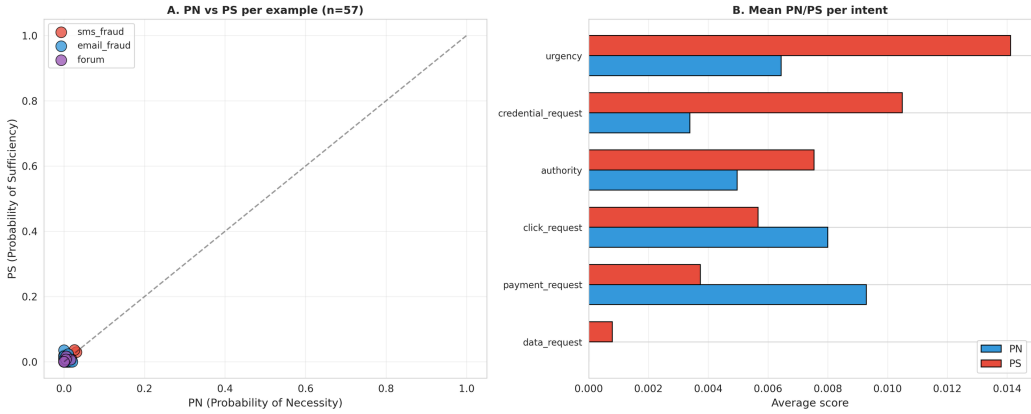


Figure 11: Gradient-based causal attribution on 42 test examples. (A) PN versus PS per example at $k = 10$, colored by intent. Points cluster near the diagonal, indicating that the top- k token set is neither overwhelmingly necessary nor sufficient. (B) Mean PN and PS per intent at $k = 10$. Explicit requests (credential_request, data_request) exhibit PN > PS; some implicit manipulations (payment_request, click_request) exhibit the reverse.

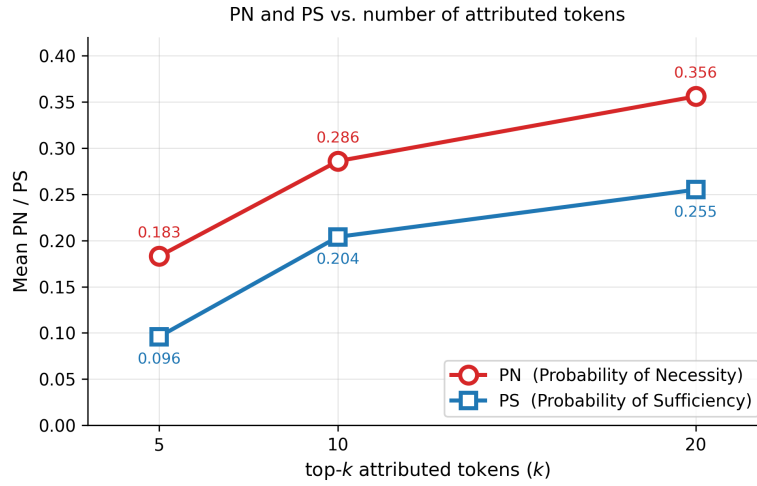


Figure 12: Mean Probability of Necessity and Probability of Sufficiency as functions of the number of attributed tokens k . Both grow monotonically, PN at a stronger rate ($0.21 \rightarrow 0.32$ for $k = 5 \rightarrow 20$). A fully distributed representation would show a flat curve; a fully localizable representation would show rapid saturation at small k . The observed sub-linear growth is consistent with a *partially localizable* representation.

6. Discussion

Why fine-tuning fails cross-channel.. The LOO experiment makes the failure mode concrete. When SMS is removed from training, the encoder has only two channels to fit: email and forum. Both look nothing like SMS. The training loss is minimised by a representation in which “SMS-like” falls outside the support of the training distribution, and the encoder defaults to the majority class within that region. This is not overfitting in the classical sense; the model has converged to a solution that is optimal on the training distribution and uninformative outside it. A canonical instance of shortcut learning (Geirhos et al., 2020): the model exploits channel-specific surface cues that predict well on the training distribution and fail to generalise.

Why training objectives fail to fix it.. The ablation study in Section 5.7 shows that training-time objectives do not achieve PCC-level invariance. BCE-only has $MI = 0.788$; adding SupCon *increases* MI to 0.828, and MMD partially returns MI to 0.676. The MMD loss itself drops during training at $\lambda = 0.3$, which confirms the term is actively aligning channels, but aggregate MI on the held-out test does not reach the PCC level. SupCon sharpens all structure in the representation — including channel-specific clustering — and MMD counteracts this without additional benefit at the chosen weight. The decisive intervention is PCC, which reduces MI from 0.676 to 0.116 without retraining and without damaging the intent signal. This matches the observation in the domain adaptation literature that MMD-based alignment alone can degrade discriminative features when the two distributions are not separable, and the broader finding that representation geometry can be manipulated more cleanly with a post-hoc linear operation than with a training-time constraint.

PN grows with k .. Classical phishing detection relies on localised triggers: a bank name, a suspicious URL, an urgency phrase. If CIC-Intents had learned such triggers *exclusively*, PN would saturate quickly at small k . If the representation were fully distributed, PN would be flat. The observed sub-linear growth ($0.21 \rightarrow 0.29 \rightarrow 0.32$) places the model in the intermediate regime: intent information is spread across a moderate number of tokens, with no single token being either necessary or sufficient on its own. Two implications follow. Token-level explanations are meaningful but not authoritative. Single-token adversarial attacks are effective only if the attacker can target a sizeable fraction of the informative tokens.

Use/mention as the dominant axis.. The leave-forum-out per-intent pattern is the clearest evidence that the difficulty is semantic rather than stylistic. Explicit requests (`credential_request`, `payment_request`, `data_request`, `click_request`) transfer at AP 0.46–0.61; implicit manipulations (`urgency`, `fear`, `impersonation`, `greed`) transfer at AP 0.05–0.27. The gap is systematic and large. We attribute it to the use/mention distinction (Gligorić et al., 2024): explicit requests are phrased almost identically whether the context is an SMS or a forum post, while implicit manipulations are refracted through the register of the channel.

Limitations.. The email corpus is LLM-generated. Real Italian phishing email distributions may differ. The Telegram corpus is a hate-speech filtered subset, used as a proxy for non-fraud content; it is not a general-purpose non-fraud corpus. The forum corpus is limited to two Italian forums. Annotation covers 950 documents, with two intents present only once in the test set. The forum test subset is treated as a case study with three independent estimates rather than a single hold-out number. We do not test on non-Italian channels or on adversarial perturbations of full documents. The reported results are single-seed; a multi-seed verification on a sub-experiment yielded a standard deviation of roughly 0.01 in macro-F1, which sits inside the reported confidence intervals. For the smallest channel, where each 5-fold CV test split contains only $n \approx 22$ documents, the run-to-run standard deviation of macro-F1 is substantially larger ($\sigma \approx 0.06$); the CV figures in Table 10 should be read with this variance in mind. The reported PCC reduction in Domain MI (−82.9%) is larger than the value reported in an earlier version of this work; the difference reflects a change in the centering protocol (per-channel means are now estimated on the training split and applied to the test split, rather than computed on the test split itself), which is the more conservative protocol.

Implications for practice.. Deploy CIC-Intents as a decision-support system. The Top-5 recall of 0.931 means a five-item shortlist per document recovers 93.1% of the true intents. Reduce the analyst’s search space by roughly a factor of 2.4 (12 intents down to 5) with minimal loss of signal. Do not deploy a single-channel model and assume it will transfer.

7. Conclusion

We studied Italian fraud detection across four channels and found that fine-tuned BERT does not transfer across them. The reason is geometric: the encoder learns a representation in which channel is linearly decodable. We proposed CIC-INTENTS, a framework that combines a multi-label intent taxonomy, a channel-weighted contrastive loss, an MMD regularizer, and Post-hoc Channel Centering (PCC). The system reaches mAP 0.721 and Top-5 recall 0.931, with balanced email and SMS performance. A controlled comparison against XLM-R and mDeBERTa shows the framework is architecture-agnostic, while the Italian-pretrained UmBERTo achieves the strongest ranking metrics despite being the smallest of the three. An ablation study confirms that PCC is the single decisive component for channel invariance, and that training-time objectives on their own do not achieve PCC-level invariance. The smallest channel is treated as a small- n case study with three independent estimates. A gradient-based attribution analysis shows that Probability of Necessity grows monotonically with the size of the attributed token set, indicating a partially localizable representation. Future work will extend the evaluation to a broader Italian corpus, to cross-language transfer, and to adversarial perturbations of the top- k tokens identified by the attribution procedure.

Reproducibility artefacts

The code, prompts, dataset splits and reproduction notebooks are publicly available. The QR code below links directly to the repository.



GitHub repository. <https://github.com/marybilik/cic-intents>

Contains all scripts, prompts, data splits and evaluation notebooks used to produce the results in this paper.

Appendix A. Reproducibility checklist

- Code, prompts and data splits: <https://github.com/marybilik/cic-intents>.
- Random seed: 42 for all experiments.
- Hardware: NVIDIA T4 (Kaggle) for training; CPU for post-hoc analyses.
- Training hyperparameters: AdamW, $\text{lr} = 5 \times 10^{-5}$, weight decay 0.01, batch 32, max sequence length 160, epochs 10.
- SupCon sanity check: on random embeddings the Multi-Label SupCon loss is ≈ 7.2 at $\tau = 0.07$; during training it stabilises around 4.0, indicating the contrastive objective is actively shaping the representation geometry.
- Annotation: Google Gemini (`gemini-flash-latest`, auto-discovered at runtime from a priority list); prompt and parser are provided in the repository. Checkpoint written every 25 documents.
- Bootstrap: 1000 resamples for all CI values.
- Split: 70/15/15 stratified by channel and intent presence for the headline model; 5-fold stratified CV and leave-forum-out for the forum channel.
- Forum small- n : three independent estimates (hold-out $n = 17$, 5-fold CV $n = 113$, leave-forum-out $n = 113$) are reported in Table 10.
- Causal attribution: gradient $\times$ input saliency, top- k token selection, $k \in \{5, 10, 20\}$, 42 test examples.
- Encoder comparison: identical framework, identical hyperparameters, identical split; checkpoints retained for UmBERTo, XLM-R-base and mDeBERTa-v3-base.

Appendix B. IFIT intent definitions

Explicit intents..

- **credential_request**: asks for login credentials, passwords, PINs, or access codes.
- **payment_request**: asks for a payment, bank transfer, or credit card details.
- **data_request**: asks for personal data (name, address, date of birth, tax code).
- **click_request**: asks the user to click a link or download an attachment.
- **call_request**: asks the user to call a phone number.

Implicit intents..

- **urgency**: creates time pressure (“act now”, “within 24 hours”).
- **authority**: invokes a trusted institution or authority figure.
- **fear**: threatens negative consequences (account blocking, legal action, loss of funds).
- **greed**: promises a reward, prize, refund, or unexpected gain.
- **impersonation**: pretends to be a known entity or person.
- **social_proof**: claims that others have already complied or benefited.
- **reciprocity**: offers something small in exchange for compliance.

Appendix C. Intent-specific vocabulary

Figure C.13 shows the intent-specific vocabulary learned by CIC-Intents. The clouds are built from lemmatized tokens of test documents whose dominant intent matches the given label, with placeholders removed. They confirm the taxonomy is grounded in distinct lexical patterns rather than a single shared vocabulary.

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